

The Utility Problem

IPT (Instrument Potential Transformer) and ICT (Instrument Current Transformer) are types of transformers that are used to measure electrical energy in a power system. They are used to reduce the voltage/current in a high-voltage circuit to a level that can be safely measured by a low-voltage instrument or meter.

Verification of electricity meter measurements is the process of checking the accuracy of the readings provided by a meter. This is done to ensure that customers are billed correctly and that the meter works properly. There are a few different methods that can be used to verify meter measurements. One of them is the comparison test - the meter's readings are compared to those of a reference meter, such as a different test instrument, to see if they match.

However, there may be cases where these IPTs and ICTs are incorrectly set and, therefore, may measure different consumption compared to the real consumption (e.g. half the consumption). This is not a random occurrence and can be seen in practice.

It is the responsibility of the utility to ensure accuracy and fair billing. If billing is not done correctly, the utility would risk reputational damage, possibly even lawsuits, or, on the other hand, would lose profits.

The Data Solutions and Methods

As mentioned above, this use case focuses on comparing two measurements. The first is the measurement using IPT or ICT. This data is then compared with another measurement device that is installed close to the IPT or ICT (it is necessary that there is no load between this device and the IPT or ICT to prevent this comparison). Any sensor that will be integrated into the Awesense Energy Transition Platform can be used as a metering device. Awesense metering equipment can also be used.

Required data:

1. Time series data from IPT or ICT.
2. Time series data from another sensor (reference measurement).

With the Awesense Platform, the user gets very easy to access to this data. If the instrument transformer and sensor data have different granularity, a predefined data aggregation function can be used to convert these source data to the same time period for easier comparison.

Each IPT or ICT has additional static information such as GPS coordinates. Thus, when a measurement discrepancy is detected, utility workers immediately know where these devices are located and can easily send a work crew to them to arrange a correction.

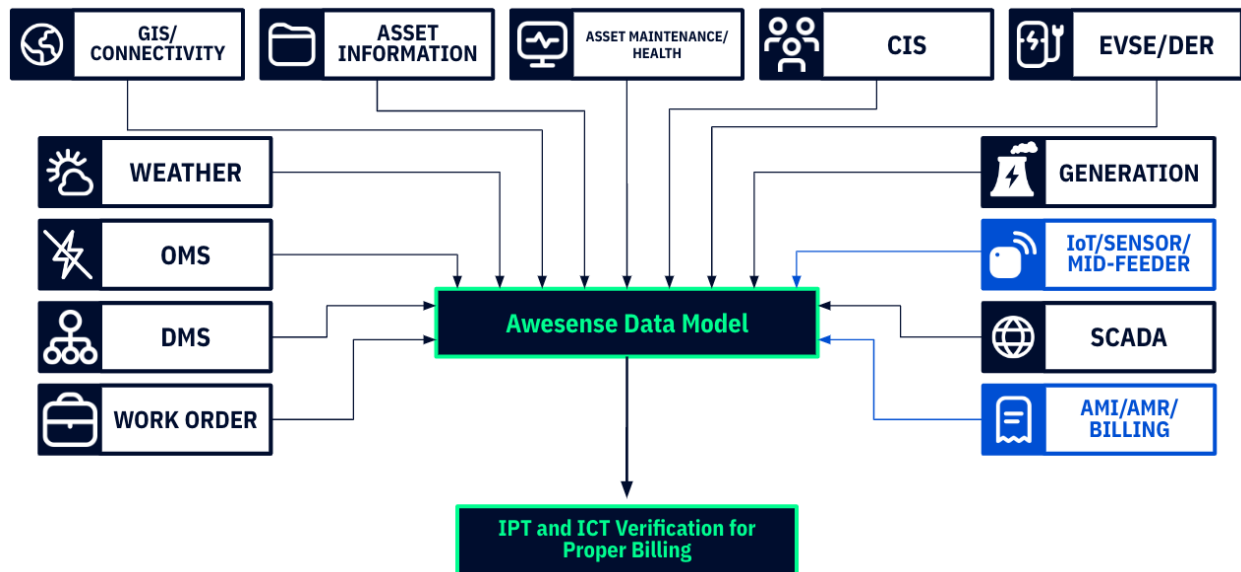
IPT and ICT Verification for Proper Billing

[UC39-01] - Awesense Energy Transition Platform Use Case



Required Data Sets

The required data for this use case are visualized in the following diagram. Data sources highlighted in [royal blue](#) are required for this use case.



Available Use Case Implementations:

Users of the Awesense Energy Transition Platform can access this use case by the following means:

1. GitHub link to the Jupyter notebook can be provided upon request.
2. TGI Energy Balance - Please speak to Awesense about access to the sandbox environment.